Optimal transport regularization of dynamic inverse problems

Silvio Fanzon

University of Graz

Based on joint works with

Kristian Bredies, Marcello Carioni and Francisco Romero

Paris, 20-21 November 2019

Plan of the Talk

1. Analytical framework for OT regularization of dynamic inverse problems (with K. Bredies)
   - *An optimal transport approach for solving dynamic inverse problems in spaces of measures.* (Preprint 2019)

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   - *An optimal transport approach for solving dynamic inverse problems in spaces of measures.* (Preprint 2019)

2. Numerical results for sparse reconstruction in spaces of measures (with K. Bredies, M. Carioni, F. Romero)
   - *On the extremal points of the ball of the Benamou-Benier energy* (Preprint 2019)
   - *A generalized conditional gradient method for dynamic inverse problems with optimal transport regularization* (In preparation)
   - *A superposition principle for the non-homogeneous continuity equation* (In preparation)

Motivation: Motion-Aware Tomographic Reconstruction

Motion on sub-acquisition time scales $\rightsquigarrow$ **artefacts** in reconstructed images

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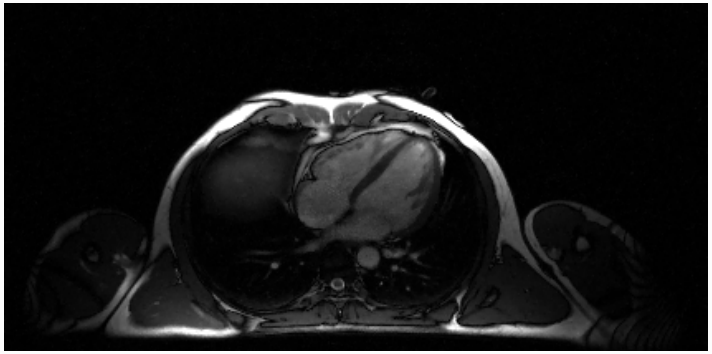

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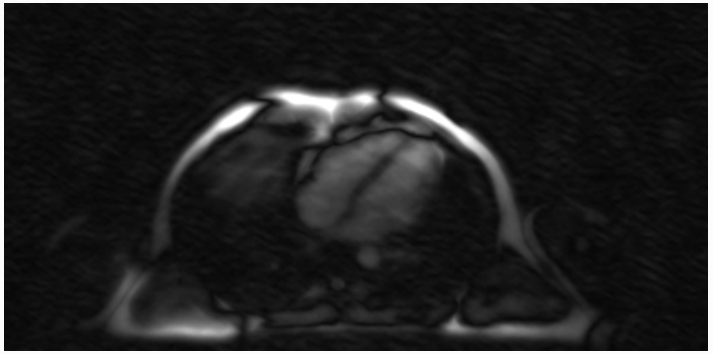

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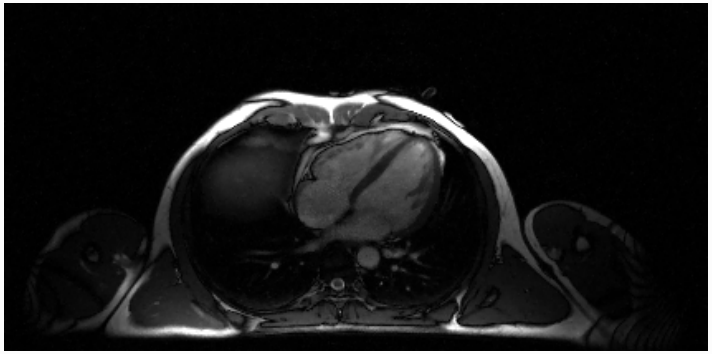

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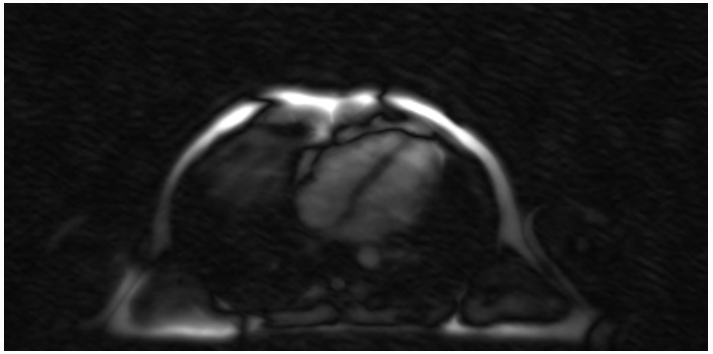

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Proposed model: optimal transport regularization for dynamic reconstruction

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$\Omega \subset \mathbb{R}^d$ bounded domain, $\rho_0, \rho_1 \in \mathcal{P}(\Omega)$, $T \colon \Omega \to \Omega$ measurable displacement

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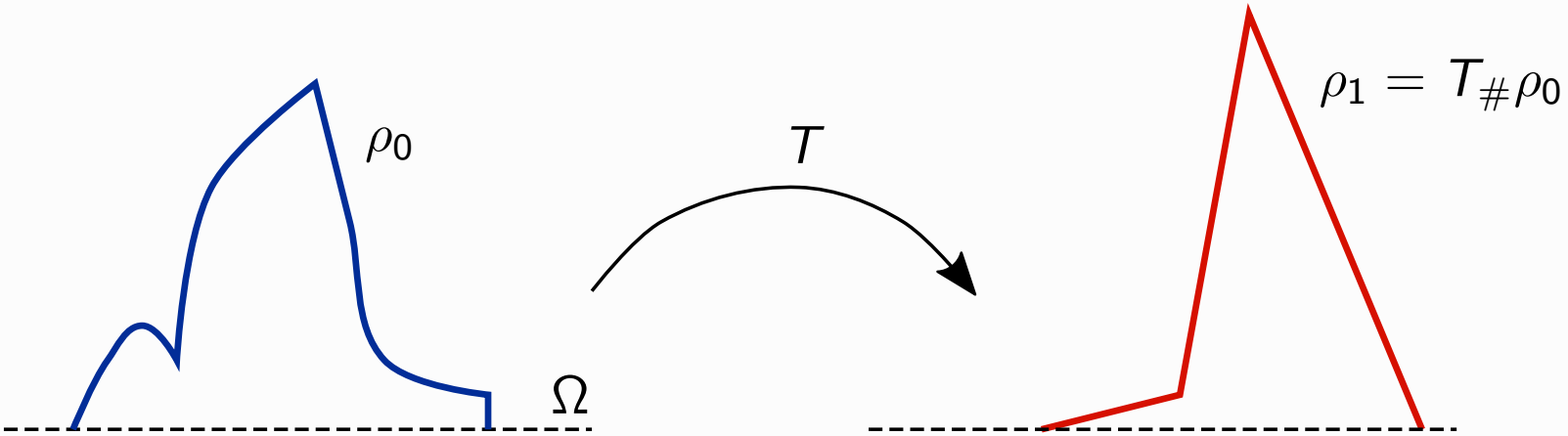

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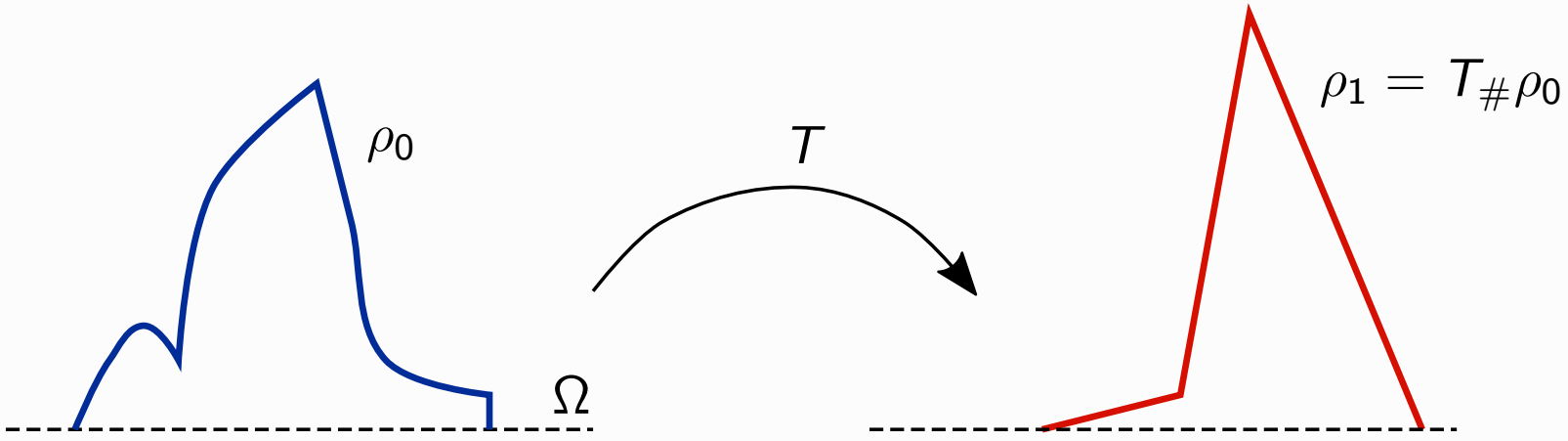

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$$c(x, y) := |x - y|^2$$

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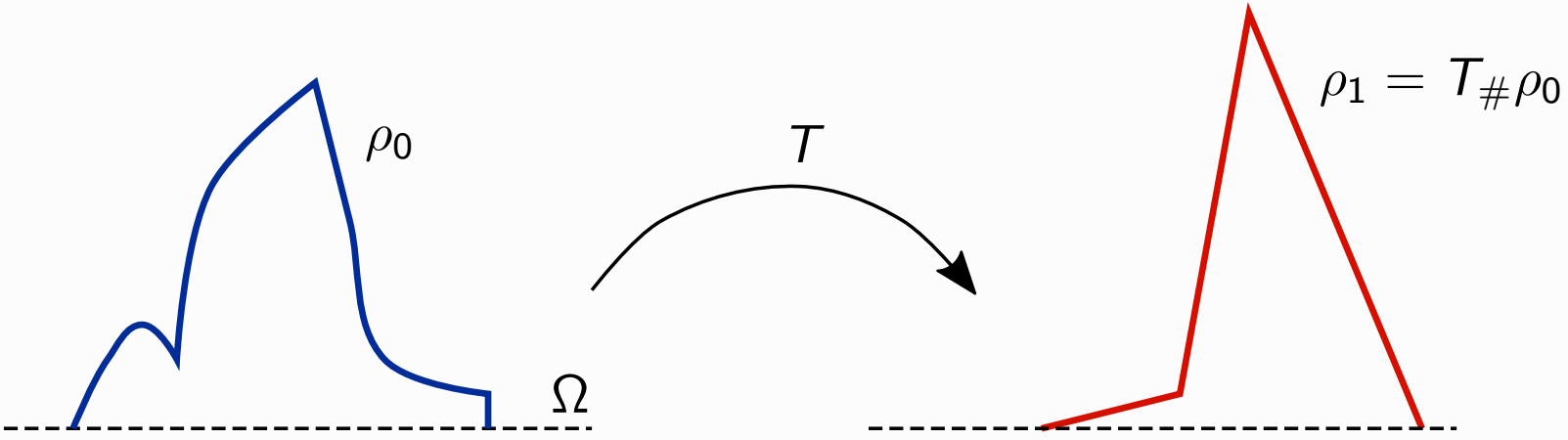

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Optimal Transport: a transport plan T solving

$$\min \left\{ \int_\Omega |T(x) - x|^2 \, d\rho_0(x) : \; T\colon \Omega \to \Omega, \; T_{\#}\rho_0 = \rho_1 \right\}$$

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- (ρ_t, v_t) solves the continuity equation with initial conditions

$$\begin{cases} \partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0 \\ \text{Initial data } \rho_0, \text{ final data } \rho_1 \end{cases} \qquad \text{(CE-IC)}$$

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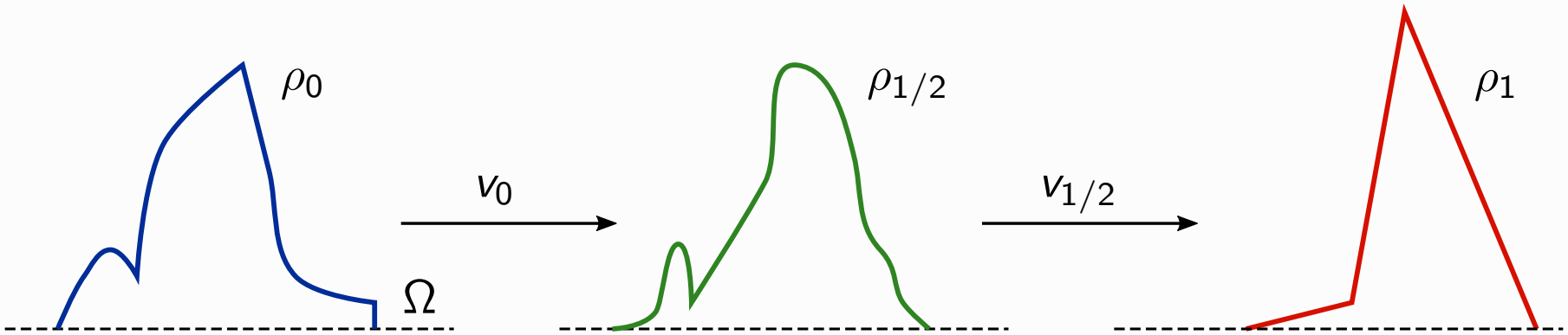

Connection and Advantages

Theorem (Benamou-Brenier '00)

$$\min_{\substack{(\rho_t, v_t) \\ \textit{solving } \text{(CE-IC)}}} \int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x) dx\, dt = \min_{\substack{T:\, \Omega \to \Omega \\ T_\# \rho_0 = \rho_1}} \int_\Omega |T(x) - x|^2 \rho_0(x)\, dx$$

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Advantages of Dynamic Formulation:

1. By introducing the momentum $m_t := \rho_t v_t$ we have

$$\int_0^1 \int_\Omega |v_t(x)|^2 \rho_t(x)\, dx\, dt = \int_0^1 \int_\Omega \frac{|m_t(x)|^2}{\rho_t(x)}\, dx\, dt$$

which is **convex** in (ρ_t, m_t).

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2. we know the full trajectory ρ_t and can recover the velocity field v_t from m_t

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Problem

Given some data $\{f_t\}_{t\in[0,1]}$ *with* $f_t \in H_t$*, find a curve of measures*

$$t \mapsto \rho_t \in \mathcal{M}(\overline{\Omega})$$

such that they solve the dynamic inverse problem

$$K_t^* \rho_t = f_t \qquad \text{for a.e.} \quad t \in [0,1]\,. \tag{P}$$

Unbalanced optimal transport regularization

Consider a triple $(\rho_t,\, v_t,\, g_t)$ with

- $t \mapsto \rho_t \in \mathcal{M}(\overline{\Omega})$ mass density (not probability measures)
- $v_t \colon (0,1) \times \overline{\Omega} \to \mathbb{R}^d$ velocity field, $g_t \colon (0,1) \times \overline{\Omega} \to \mathbb{R}$ growth rate

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$$\underbrace{\frac{1}{2}\int_0^1 \|K_t^*\rho_t - f_t\|_{H_t}^2 \, dt}_{\text{Fidelity Term}} + \frac{\alpha}{2}\underbrace{\int_0^1 \int_{\overline{\Omega}} |v_t(x)|^2 + |g_t(x)|^2 \, d\rho_t(x) dt}_{\text{Optimal Transport Regularizer}} + \beta \underbrace{\int_0^1 \|\rho_t\| \, dt}_{\text{TV Regularizer}}$$

$$\text{s.t.} \quad \partial_t \rho + \operatorname{div}(\rho v_t) = \rho g_t \qquad \text{(Continuity Equation - No IC)}$$

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Chizat, Peyré, Schmitzer, Vialard (Found. of Comp. Math. '18, JFA '18)

Liero, Mielke, Savaré (Inv. Math. '18)

Formal definition of the Unbalanced OT Energy

Set $X := (0,1) \times \overline{\Omega}$ and consider triples $(\rho, m, \mu) \in \mathcal{M}(X) \times \mathcal{M}(X)^d \times \mathcal{M}(X)$

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Define the convex, 1-homogeneous functional

$$B(\rho, m, \mu) := \int_X \Psi\left(\frac{d\rho}{d\lambda}, \frac{dm}{d\lambda}, \frac{d\mu}{d\lambda}\right) d\lambda$$

where $\lambda \in \mathcal{M}^+(X)$ is such that $\rho, m, \mu \ll \lambda$ and

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Proposition

B is weak lower-semicontinuous. If $B(\rho, m, \mu) < +\infty$ and $\partial_t \rho + \operatorname{div} m = \mu$ then*

- $\rho = dt \otimes \rho_t$ *for a weak*-continuous curve* $t \mapsto \rho_t \in \mathcal{M}^+(\overline{\Omega})$
- $m = \rho v_t$ *for some velocity field* $v_t \colon (0,1) \times \overline{\Omega} \to \mathbb{R}^d$
- $\mu = \rho g_t$ *for some growth rate* $g_t \colon (0,1) \times \overline{\Omega} \to \mathbb{R}$

$$B(\rho, m, \mu) = \int_0^1 \int_{\overline{\Omega}} |v_t(x)|^2 + |g_t(x)|^2 \, d\rho_t(x) \, dt$$

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Step Functions: a map $\varphi \colon [0,1] \to D$ is a step function if

$$\varphi_t = \sum_{j=1}^{N} \chi_{E_j}(t)\, \varphi_j$$

for $\varphi_j \in D$, $E_j \subset [0,1]$ measurable, $N \in \mathbb{N}$.

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Strong Measurability: a map $f \colon [0,1] \to \cup_t H_t$ with $f_t \in H_t$ is str. meas. if $\exists\ \varphi^n \colon [0,1] \to D$ step functions s.t.

$$\lim_n \|i_t\varphi_t^n - f_t\|_{H_t} = 0 \quad \text{for a.e.} \quad t \in (0,1)$$

Sampling spaces 2/3

Integrability: a str. meas. map $f\colon [0,1] \to \cup_t H_t$ with $f_t \in H_t$ is integrable if $\exists\ \varphi^n\colon [0,1] \to D$ step functions s.t.

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Theorem (SF, Bredies '19)

Let $f\colon [0,1] \to \cup_t H_t$ be strongly measurable. Then f is **integrable** *iff*

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Note: it is possible to show the Theorem after introducing suitable notions of weakly measurable and of separably valued maps $f\colon [0,1] \to \cup_t H_t$, in a way that a version of Pettis Theorem holds.

Sampling spaces 3/3

Definition (Data space)

$$L^2_H = \left\{ f : [0,1] \to \cup_t H_t : \ f_t \in H_t, \ f \text{ strongly meas } , \ \int_0^1 \|f_t\|^2_{H_t} \, dt < \infty \right\}$$

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Sampling spaces 3/3

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$$L^2_H = \left\{ f\colon [0,1] \to \cup_t H_t \,:\; f_t \in H_t,\, f \text{ strongly meas } ,\; \int_0^1 \|f_t\|^2_{H_t}\, dt < \infty \right\}$$

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Forward operators and Regularized Problem

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Definition (Regularization)

Let $f \in L^2_H$ be some data. For $(\rho, m, \mu) \in \mathcal{M}(X) \times \mathcal{M}(X)^d \times \mathcal{M}(X)$ set

$$T_{\alpha,\beta}(\rho, m, \mu) := \frac{1}{2} \int_0^1 \|K_t^* \rho_t - f_t\|^2_{H_t} \, dt + \alpha \, B(\rho, m, \mu) + \beta \, \|\rho\|_{\mathcal{M}(X)}$$

if $\partial_t \rho + \operatorname{div} m = \mu$, and $T_{\alpha,\beta}(\rho, m, \mu) = +\infty$ else.

Existence & Stability

Theorem (SF, Bredies '19)

Assume (H)-(K) and let $f \in L^2_H$. Then

$$\min_{(\rho,m,\mu)\in\mathcal{M}(X)^{d+2}} T_{\alpha,\beta}(\rho, m, \mu) \tag{MIN}$$

admits a solution. If K_t^* is injective for a.e. t, then the solution is unique.

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$$(\rho^n, m^n, \mu^n) \overset{*}{\rightharpoonup} (\rho^\dagger, m^\dagger, \mu^\dagger) \;\; \text{in} \;\; \mathcal{M}(X)^{d+2}, \quad K_t^*\rho_t^\dagger = f_t^\dagger \;\; \text{for all} \;\; t \in [0,1]$$

$$(\rho^\dagger, m^\dagger, \mu^\dagger) \in \arg\min \; \alpha^* B(\rho,m,\mu) + \beta^* \|\rho\|_{\mathcal{M}(X)}, \quad \exists\, \alpha^*, \beta^* \geq 1$$

Variational reconstruction for undersampled MRI 1/2

- $\Omega = [0,1]^2$ image frame, $t \mapsto \rho_t \in \mathcal{M}(\overline{\Omega})$ proton density

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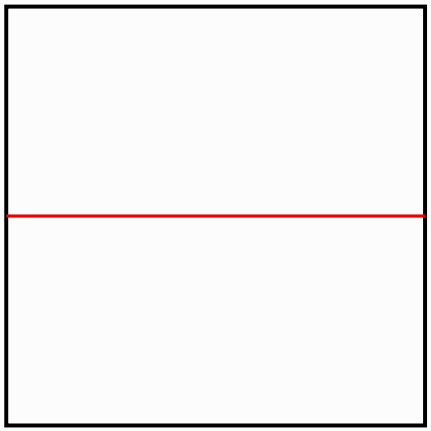

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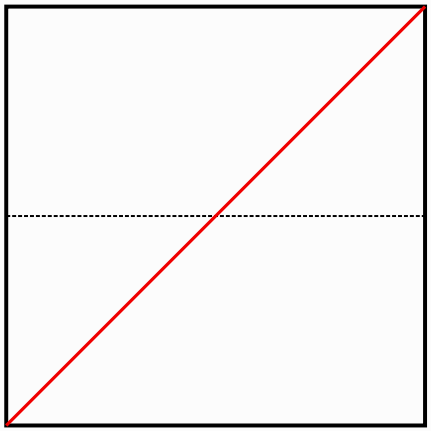

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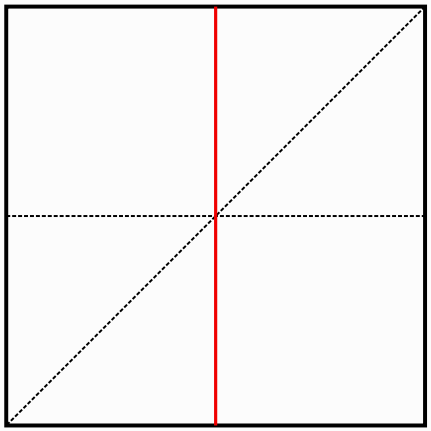

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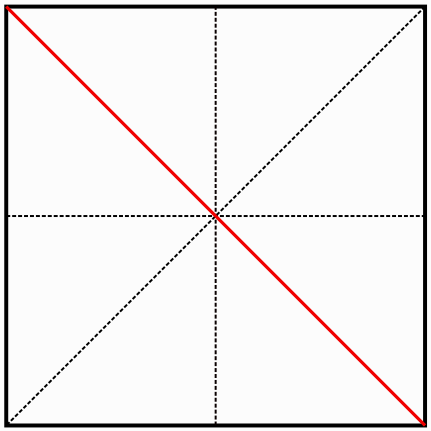

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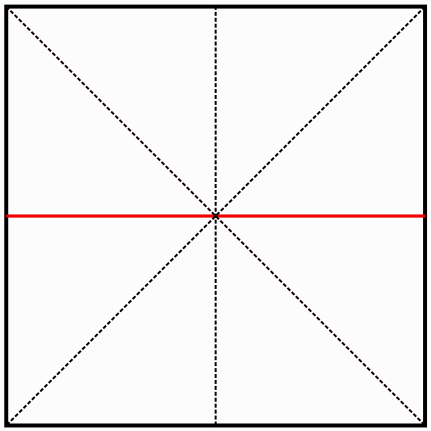

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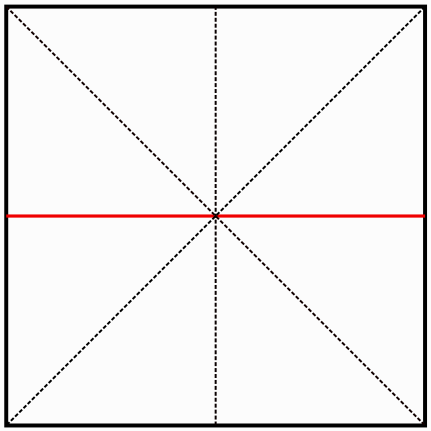

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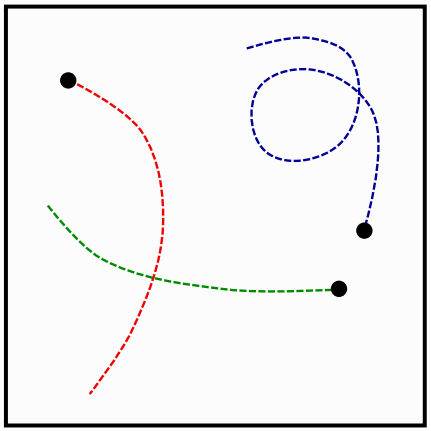

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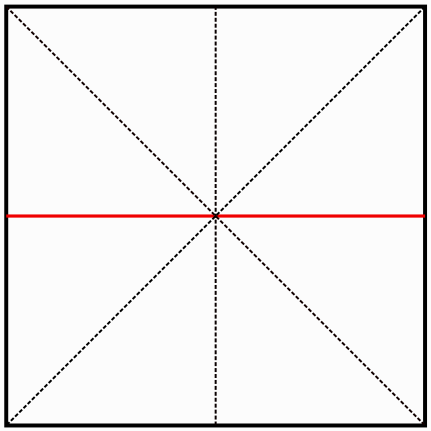

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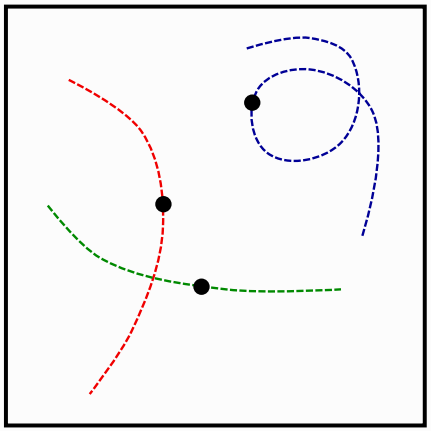

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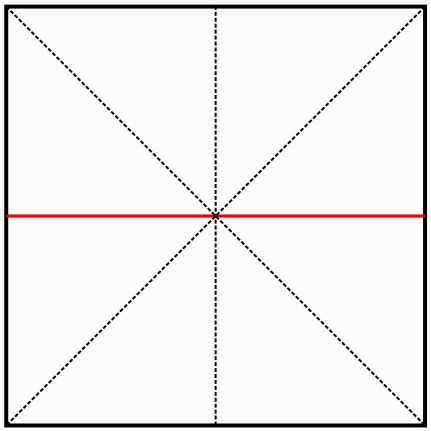

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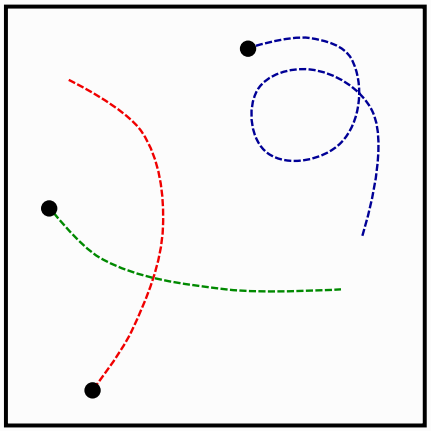

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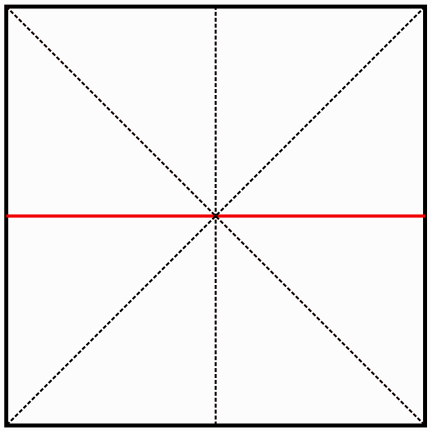

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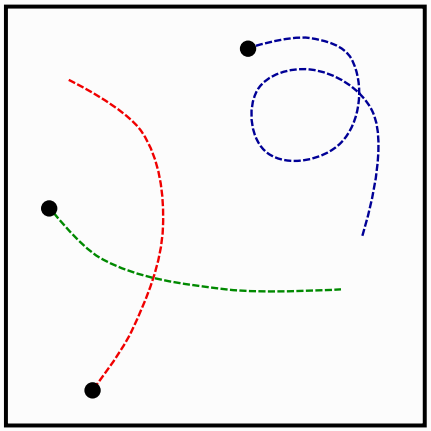

$\sigma_t = \mathcal{H}^0 \llcorner P_t$

- $K_t^* \colon \mathcal{M}(\overline{\Omega}) \to H_t$ masked Fourier transform

$$K_t^* \rho := (\mathfrak{F}(c_1 \rho), \ldots, \mathfrak{F}(c_N \rho))$$

with $c_j \in C_0(\mathbb{R}^2; \mathbb{C})$ coil sensitivities (accounting for phase inhomogeneities)

Variational reconstruction for undersampled MRI 2/2

(M): Assume that the family $\sigma_t \in \mathcal{M}^+(\mathbb{R}^2)$ satisfies:

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Theorem (SF, Bredies '19)

Assume (M). Let $\alpha, \beta, \delta > 0$, $f \in L^2_H$ and $c \in C_0(\mathbb{R}^2;\mathbb{C}^N)$. Then

$$\min_{\substack{(\rho,m,\mu)\in\mathcal{M}(X)^4 \\ \partial_t\rho+\operatorname{div} m=\mu}} \frac{1}{2}\sum_{j=1}^{N}\int_0^1 \|\mathfrak{F}(c_j\rho_t) - f_t\|^2_{L^2_{\sigma_t}}\, dt + \alpha B_\delta(\rho, m, \mu) + \beta\,\|\rho\|$$

admits a solution (ρ, m, μ) with

- $\rho = dt \otimes \rho_t$ *with* $t \mapsto \rho_t$ *weak* continuous*
- $m = \rho v$ *for some velocity* $v\colon (0,1)\times\overline{\Omega} \to \mathbb{R}^2$
- $\mu = \rho g$ *for some growth rate* $g\colon (0,1)\times\overline{\Omega} \to \mathbb{R}^2$

Numerical Results for Benamou-Brenier regularizer

Bibliography:

- SF, Bredies, Carioni, Romero - *On the extremal points of the ball of the Benamou-Benier energy* (Preprint 2019)
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Let $\Omega \subset \mathbb{R}^d$ be open bounded, $X := (0,1) \times \overline{\Omega}$. Consider the Benamou-Brenier energy

$$B(\rho, m) := \int_X \Psi\left(\frac{d\rho}{d\lambda}, \frac{dm}{d\lambda}\right) d\lambda$$

where $\lambda \in \mathcal{M}^+(X)$ is such that $\rho, m \ll \lambda$ and

$$\Psi(t,x) := \frac{|x|^2}{2t} \quad \text{if} \quad t > 0, \quad \Psi = +\infty \text{ else}$$

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Recall: if $\rho = dt \otimes \rho_t$ and $m = v\rho$

$$B(\rho, m) := \frac{\alpha}{2} \int_0^1 |v(t,x)|^2 \, d\rho_t(x) \, dt$$

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Definition (Regularizer)

Let $\alpha, \beta > 0$. For $(\rho, m) \in \mathcal{M}(X) \times \mathcal{M}(X)^d$ we set

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Problem

Given $f \in L^2([0, 1]; H)$ compute a minimizer $(\rho, m) \in \mathcal{M}(X)^{d+1}$ for

$$T_{\alpha,\beta}(\rho, m) := \frac{1}{2} \int_0^1 \|K_t^* \rho_t - f_t\|_{H_t}^2 + J_{\alpha,\beta}(\rho, m)$$

Extremal points of $J_{\alpha,\beta}$

Consider the convex unit ball of $J_{\alpha,\beta}$

$$C := \left\{(\rho, m) \in \mathcal{M}(X) \times \mathcal{M}(X)^d : \; J_{\alpha,\beta}(\rho, m) \leq 1\right\}$$

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For $\gamma \in \mathrm{AC}^2([0,1];\overline{\Omega})$ define the measures $\rho_\gamma \in \mathcal{M}(X)$, $m_\gamma \in \mathcal{M}(X)^d$ as

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Theorem (Fanzon, Bredies, Carioni, Romero '19)

The extremal points of C are characterized by

$$\mathrm{Extr}(C) = \{(0,0)\} \cup \mathcal{C}$$

where

$$\mathcal{C} := \left\{(\rho_\gamma, m_\gamma) : \; \gamma \in \mathrm{AC}^2([0,1];\overline{\Omega})\right\}$$

Idea of the proof - $\{0\} \cup \mathcal{C} \subset \mathrm{Extr}(C)$

The case $(\rho, m) = (0, 0)$ is trivial.

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- $J_{\alpha,\beta}(\rho_\gamma, m_\gamma) = 1$: Take $\lambda := \rho_\gamma$ and recall that $\Psi(t,x) = |x|^2/2t$,

$$J_{\alpha,\beta}(\rho_\gamma, m_\gamma) = \alpha \int_0^1 \int_\Omega \Psi\left(\frac{d\rho_\gamma}{d\lambda}, \frac{dm_\lambda}{d\lambda}\right) \, d\lambda + \beta \, \|\rho\|_{\mathcal{M}(X)}$$
$$= a_\gamma \left(\alpha \int_0^1 \int_\Omega \Psi\left(1, \dot{\gamma}(t)\right) \, d\delta_{\gamma(t)} \, dt + \beta \right) = a_\gamma \, a_\gamma^{-1} = 1$$

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- Since $J_{\alpha,\beta}(\rho^j, m^j) = 1$ then $a^j = a_\gamma$. Hence $(\rho^j, m^j) = (\rho_\gamma, m_\gamma)$

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Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Consider

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There exists $\sigma \in P(\Gamma)$ such that

$$\int_{\overline{\Omega}} \varphi(x)\, d\rho_t(x) = \int_\Gamma \varphi(\gamma(t))\, d\sigma(\gamma) \qquad \text{for all} \quad \varphi \in C(\overline{\Omega}),\ t \in [0,1]$$

Moreover σ is concentrated on curves $\gamma \in AC^2([0,1];\overline{\Omega})$ such that

$$\dot{\gamma}(t) = v(t, \gamma(t))$$

L. Ambrosio (Inv. Math. '04) for $\Omega = \mathbb{R}^d$

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- Assume (ρ, m) is an extremal point: Since $J_{\alpha,\beta}(\rho, m) \leq 1$ we can apply the Theorem and represent ρ via (F). Extremality forces

$$\sigma = \delta_{\gamma^*} \quad \text{for some} \quad \gamma^* \in AC^2([0,1]; \overline{\Omega}) \quad \overset{(F)}{\Longrightarrow} \quad \text{Thesis}$$

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Inverse problem: for $(f_1, \ldots, f_N) \in \mathcal{H}$ find a curve $t \mapsto \rho_t \in \mathcal{M}(\overline{\Omega})$ such that

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Regularization: we regularize via

$$T_{\alpha,\beta}(\rho, m) := \frac{1}{2} \sum_{i=1}^{N} \|K_i \rho_{t_i} - f_i\|_{H_i}^2 + \alpha B(\rho, m) + \beta \|\rho\|_{\mathcal{M}(X)}$$

Sparse minimizers

Theorem (Fanzon, Bredies, Carioni, Romero '19)

The minimization problem

$$\min_{(\rho,m)\in\mathcal{M}(X)^{d+1}} \frac{1}{2}\sum_{i=1}^{N} \|K_i\rho_{t_i} - f_i\|_{H_i}^2 + \alpha B(\rho,m) + \beta\,\|\rho\|_{\mathcal{M}(X)}$$

admits a sparse minimizer of the form

$$(\rho^*, m^*) = \sum_{i=1}^{p} c_i\,(\rho_{\gamma_i}, m_\gamma)$$

where $c_i > 0$, $\gamma_i \in \mathrm{AC}^2([0,1];\overline{\Omega})$ *and* $p \leq \dim \mathcal{H}$.

K. Bredies, M. Carioni (Calc. Var. PDEs '19)

C. Boyer, A. Chambolle, Y. De Castro, V. Duval, F. De Gournay, P. Weiss (SIAM Opt. '19)

Sparse reconstruction (time continuous case)

Definition (Regularizer)

Let $\alpha, \beta > 0$. For $(\rho, m) \in \mathcal{M}(X) \times \mathcal{M}(X)^d$ we set

$$J_{\alpha,\beta}(\rho, m) := \begin{cases} \alpha B(\rho, m) + \beta \|\rho\|_{\mathcal{M}(X)} & \text{if } \partial_t \rho + \operatorname{div} m = 0 \\ +\infty & \text{otherwise} \end{cases}$$

For $t \in [0, 1]$ assume given:

- H_t family of Hilbert spaces satisfying (H)
- $K_t^* \colon \mathcal{M}(\overline{\Omega}) \to H_t$ linear continuous operators satisfying (K)

Problem

Given $f \in L^2([0, 1]; H)$ compute a minimizer $(\rho, m) \in \mathcal{M}(X)^{d+1} \times \mathcal{M}(X)^d$ for

$$T_{\alpha,\beta}(\rho, m) := \frac{1}{2} \int_0^1 \|K_t^* \rho_t - f_t\|_{H_t}^2 + J_{\alpha,\beta}(\rho, m)$$

Generalized conditional gradient method 1/2

First one can replace

$$\min_{(\rho,m)} T_{\alpha,\beta}(\rho,m) := \frac{1}{2}\int_0^1 \|K_t^*\rho_t - f_t\|_{H_t}^2 + J_{\alpha,\beta}(\rho,m)$$

by the equivalent problem

$$\min_{(\rho,m)} \tilde{T}_{\alpha,\beta}(\rho,m) := \frac{1}{2}\int_0^1 \|K_t^*\rho_t - f_t\|_{H_t}^2 + \varphi(J_{\alpha,\beta}(\rho,m)) \tag{P}$$

where $\varphi\colon \mathbb{R} \to [0,\infty]$ is for example

$$\varphi(t) := t + \chi_{\{s \leq M_0\}}(t)\,, \quad M_0 := \frac{1}{2}\int_0^1 \|f_t\|_{H_t}^2 \, dt$$

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Then one approximates (P) by linearizing the quadratic term around $(\tilde{\rho},\tilde{m})$

$$\min_{(\rho,m)} -\int_0^1 \langle \rho_t, w_t\rangle_{\mathcal{M}(\overline{\Omega}),C(\overline{\Omega})}\, dt + \varphi(J_{\alpha,\beta}(\rho,m))\,, \quad w_t := -K_t(K_t^*\tilde{\rho}_t - f_t)$$

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Consider the convex unit ball of $J_{\alpha,\beta}$

$$C := \left\{(\rho, m) \in \mathcal{M}(X)^{d+1} : \; J_{\alpha,\beta}(\rho, m) \leq 1\right\}$$

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Then there exists a solution $(\rho^*, m^*) \in \mathrm{Extr}(C)$ to the problem

$$\min_{(\rho,m)\in C} - \int_0^1 \langle \rho_t, w_t \rangle_{\mathcal{M}(\overline{\Omega}), C(\overline{\Omega})} \, dt \tag{L}$$

Moreover there exists $M \geq 0$ such that $(M\rho^*, Mm^*)$ is a solution to

$$\min_{(\rho,m)} - \int_0^1 \langle \rho_t, w_t \rangle_{\mathcal{M}(\overline{\Omega}), C(\overline{\Omega})} \, dt + \varphi(J_{\alpha,\beta}(\rho, m))$$

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2. (Optimization) Solve the quadratic problem

$$\bar{C} = (\bar{c}_j)_j \in \underset{c_j \geq 0}{\arg\min}\ T_{\alpha,\beta}\left(\rho^{n+1/2}, m^{n+1/2}\right)$$

   Define

$$(\rho^{n+1}, m^{n+1}) := \sum_j \bar{c}_j \left(\rho_{\gamma_j}, m_{\gamma_j}\right)$$

Convergence

Define the functional distance

$$r(\rho, m) := T_{\alpha,\beta}(\rho, m) - \min T_{\alpha,\beta}$$

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Let $f \in L^2_H$, $\alpha, \beta > 0$ and $(\rho^n, m^n) \in \mathcal{M}(X)^{d+1}$ be the sequence in the Algorithm

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$$r(\rho^n, m^n) \leq \frac{C}{n} \tag{B}$$

where $C > 0$ depends only on f, α, β

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Improving convergence: In the simulations we obtain linear convergence. Therefore one could expect that (B) can be improved (current proof does not take advantage of coefficients optimization step)

Simulations 1/4

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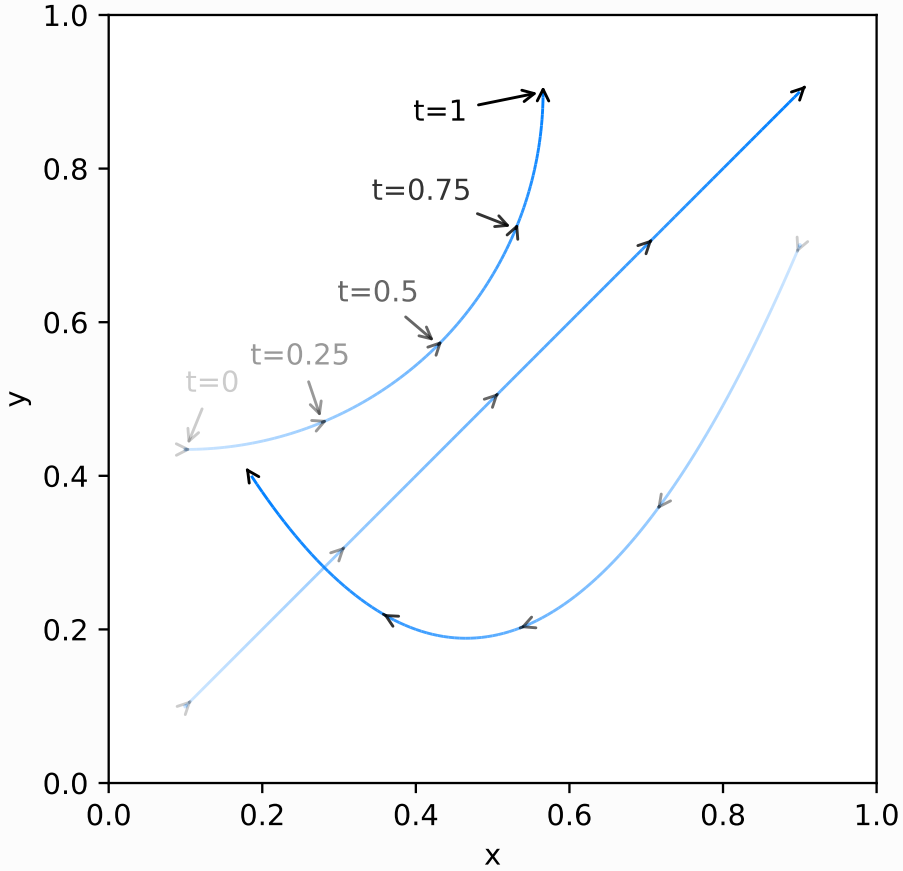
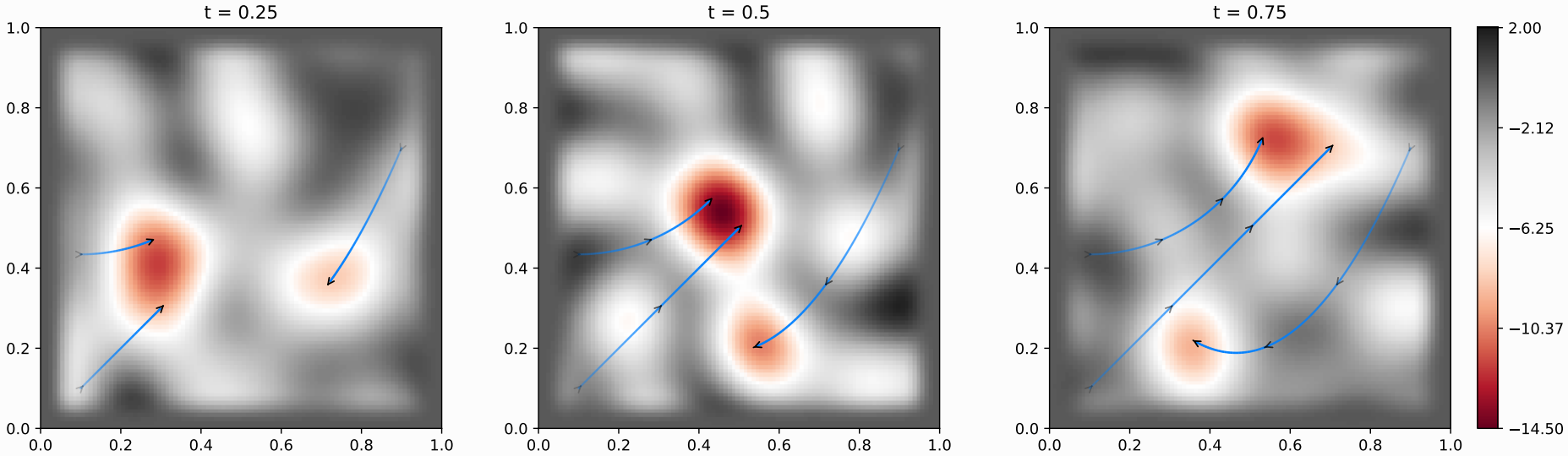
- Data is $f_t = K_t^* \rho_t$ where ρ_t are the curves in picture (no noise)



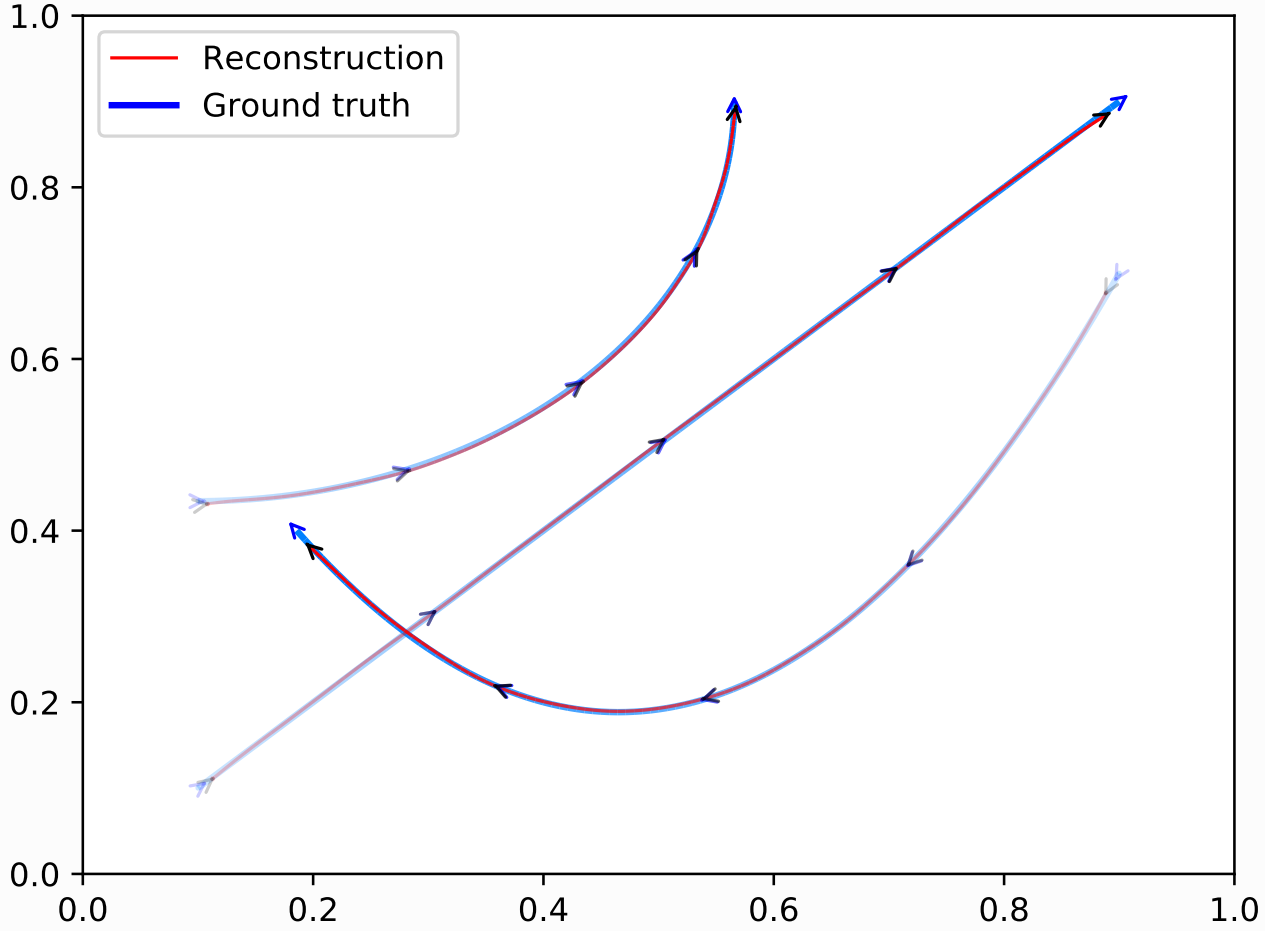
Simulations 2/4



Dual variable first iteration $w_t := -K_t f_t$

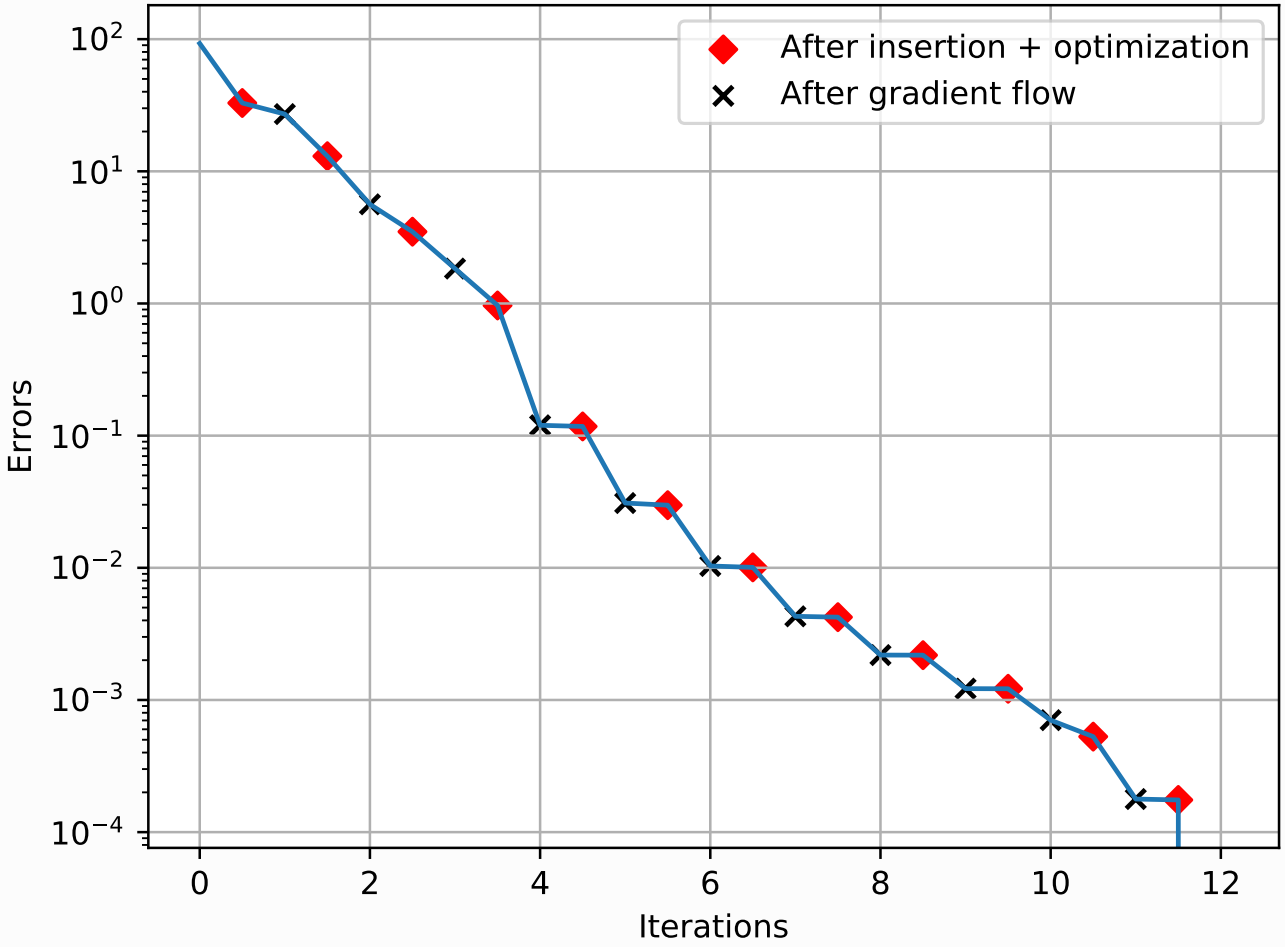
(Recall that $w_t \in C(\overline{\Omega})$ for each $t \in [0,1]$)

Simulations 3/4



Reconstructed trajectories

Simulations 4/4



Convergence plot: exhibits linear rate

$$\text{Error} = T_{\alpha,\beta}(\rho^n, m^n) - T_{\alpha,\beta}(\rho^{n+1}, m^{n+1})$$

Further directions: unbalanced OT case

Consider the regularizer

$$J_{\alpha,\beta}(\rho, m, \mu) := \alpha \int_X \Psi\left(\frac{d\rho}{d\lambda}, \frac{dm}{d\lambda}, \frac{d\mu}{d\lambda}\right) d\lambda + \beta \|\rho\|_{\mathcal{M}(X)} \quad \text{if} \quad \partial_t \rho + \operatorname{div} m = \mu$$

where $\lambda \in \mathcal{M}^+(X)$ is such that $\rho, m, \mu \ll \lambda$ and $\Psi(t, x, y) := \frac{|x|^2 + y^2}{2t}$ if $t > 0$

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Key ingredients:

- Characterization of the extremal points of $J_{\alpha,\beta}$, which are of the form

$$\rho = h(t)\, dt \otimes \delta_{\gamma(t)}\,, \quad m = \dot{\gamma}\rho\,, \quad \mu = \frac{\dot{h}}{h}\,\rho$$

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- Characterization is based on a superposition principle for $\partial_t \rho + \operatorname{div} m = \mu$

SF, Bredies, Carioni, Romero - A superposition principle for the non-homogeneous continuity equation (In preparation)

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Thank You!